TECHNICAL SECURITY ASSESSMENT

DEATHNOTE

VulnHub Boot-to-Root Assessment

Penetration Testing Report

TARGET IP	192.168.56.111
HOSTNAME	deathnote.vuln
ASSESSMENT TYPE	Controlled, intentionally vulnerable virtual machine
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REPORT DATE	8 November 2026

Prepared for technical learning and portfolio documentation.
All testing was limited to the authorised lab environment.

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1 Executive Summary

This report documents a controlled penetration-testing assessment of the Deathnote vulnerable virtual machine. The assessment followed a web-led enumeration path: the exposed web service and WordPress content were inspected, clues in `robots.txt` and an image file were followed, encoded data was analysed, SSH credentials were recovered, and local privilege escalation was completed.

The supplied evidence identifies the target as `192.168.56.111` (`deathnote.vuln`), running Apache on Debian and hosting a WordPress installation. The evidence further shows SSH access as the account displayed as `l` (lowercase L), subsequent access to the account `kira`, and root-level command execution through an unrestricted sudo rule assigned to `kira`.

> **Assessment outcome**
> The machine was fully compromised in the lab environment. The clearest demonstrated escalation path was the unrestricted sudo permission `(ALL : ALL) ALL` assigned to `kira`, followed by a root shell and access to `/root/root.txt`.

The supplied screenshots do not include the initial Nmap output or a complete transcript of every intermediate command. This report therefore separates directly visible evidence from conclusions and does not invent missing scan results or undocumented steps.

2 Assessment Scope and Environment

Target IP	`192.168.56.111`
Target hostname	`deathnote.vuln`
Operating system evidence	Debian GNU/Linux; kernel `4.19.0-17-amd64`; x86_64 (SSH login banner)
Web service	Apache `2.4.38 (Debian)` on HTTP port 80
Other confirmed service	SSH on port 22
Assessment type	Local, intentionally vulnerable boot-to-root virtual machine
Tools visible in evidence	WPScan output, Hydra, SSH client, curl, hexadecimal/Base64 decoding and a Brainfuck interpreter

Testing was restricted to the assigned lab machine. No production system or third-party host was assessed.

3 Attack Path Overview

1. **Web reconnaissance:** Identified the Apache-hosted site and WordPress installation under `/wordpress`.
2. **Content discovery:** WordPress enumeration exposed useful paths, including the login page and a browsable uploads directory. `robots.txt` pointed to `/important.jpg`.
3. **Clue analysis:** The image and website hint led to references to `user.txt` and `notes.txt`. The

user file contained Brainfuck-encoded content.

4. **Credential clue recovery:** A hexadecimal value was converted to text and then Base64-decoded, revealing the clue `passwd : kiraisevil`.
5. **SSH access:** Hydra output reported a successful SSH credential pair for account `l` (lowercase L) and password `death4me`.
6. **Local investigation:** Subsequent evidence shows access to `kira`, inspection of `kira.txt`, and investigation of `/opt/fake-notebook-rule` and `case.wav`.
7. **Privilege escalation:** The `kira` account had unrestricted sudo privileges. Running `sudo su` produced a root shell, after which `/root/root.txt` was accessed.

4 Detailed Findings

4.1 Web Service and WordPress Enumeration

The screenshots identify Apache `2.4.38 (Debian)` on port 80 and a WordPress installation under `/wordpress`. WordPress enumeration output reports the login page, a browsable `wp-content/uploads` directory, and other exposed WordPress paths. Directory browsing can disclose uploaded files and implementation details that support further reconnaissance.

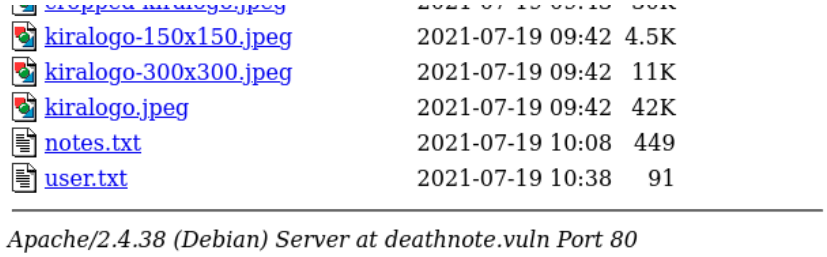


Figure 1: Web directory discovery and WordPress-related content.

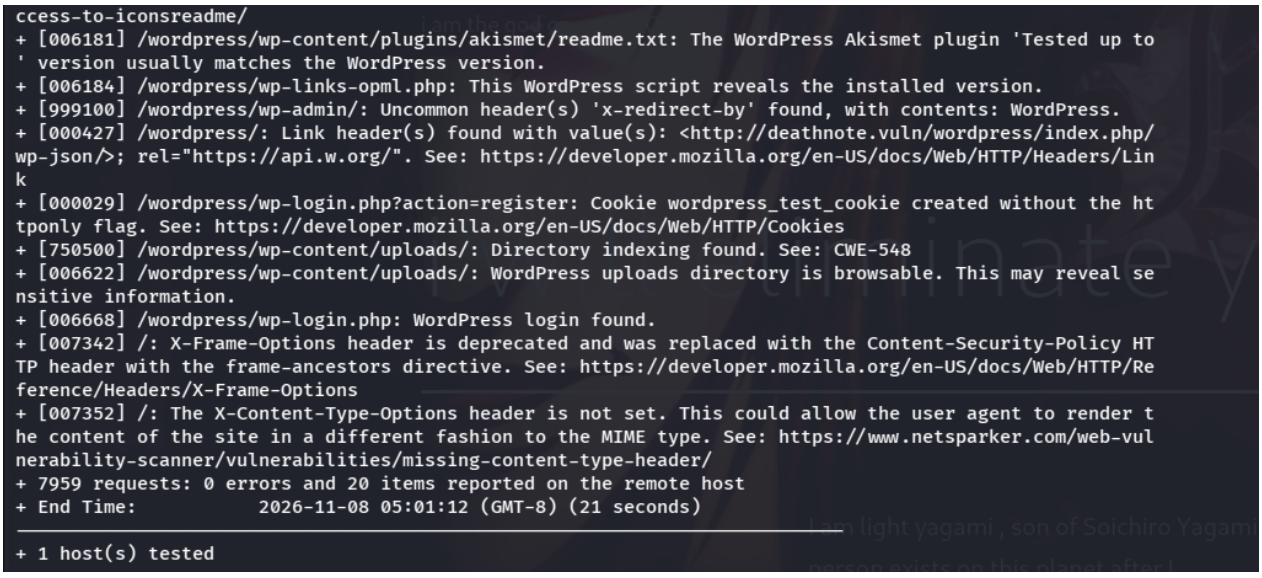


Figure 2: WordPress enumeration output identifying exposed paths, including the login page and uploads directory.

4.2 Information Disclosure Through `robots.txt` and Image Content

The `robots.txt` file contained a message directing investigation to `/important.jpg`. The retrieved image content included a note attributed to Soichiro Yagami, referenced `user.txt`, and indicated that a password clue was available in the site's hint section. The hint page separately instructed the tester to locate a `notes.txt` file on the server.

```
┌──(root㉿kali)-[/home/kali/evidence_dn]
└─# curl http://192.168.56.111/robots.txt
fuck it my dad
added hint on /important.jpg

ryuk please delete it
```

Figure 3: The robots.txt clue directing investigation to important.jpg.

```
┌──(root㉿kali)-[/home/kali/evidence_dn]
└─# curl http://192.168.56.111/important.jpg
i am Soichiro Yagami, light's father
i have a doubt if L is true about the assumption that light is kira

i can only help you by giving something important

login username : user.txt
i don't know the password.
find it by yourself
but i think it is in the hint section of site
```

Figure 4: Content recovered from important.jpg, including references to user.txt and the password hint.

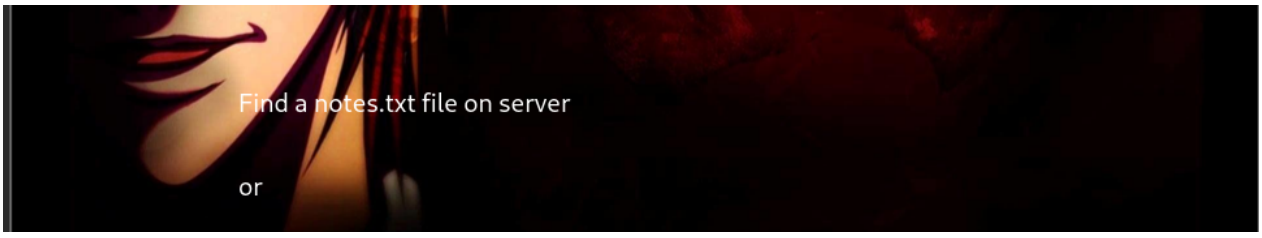


Figure 5: Website hint directing investigation toward notes.txt.

4.3 Encoded Data Analysis

The `user.txt` file contained Brainfuck-encoded text. The supplied decoder screenshot shows a decoded message referencing Kira. Another clue was represented as hexadecimal bytes; converting those bytes to text produced a Base64 string. Decoding that string yielded the text `passwd : kiraisevil`.

Credential handling note
Recovered credentials are included here only to explain the lab's attack path. Redact passwords, session material and challenge secrets before publishing this report or its screenshots to a public repository.



Figure 6: The user.txt file containing Brainfuck-encoded text.

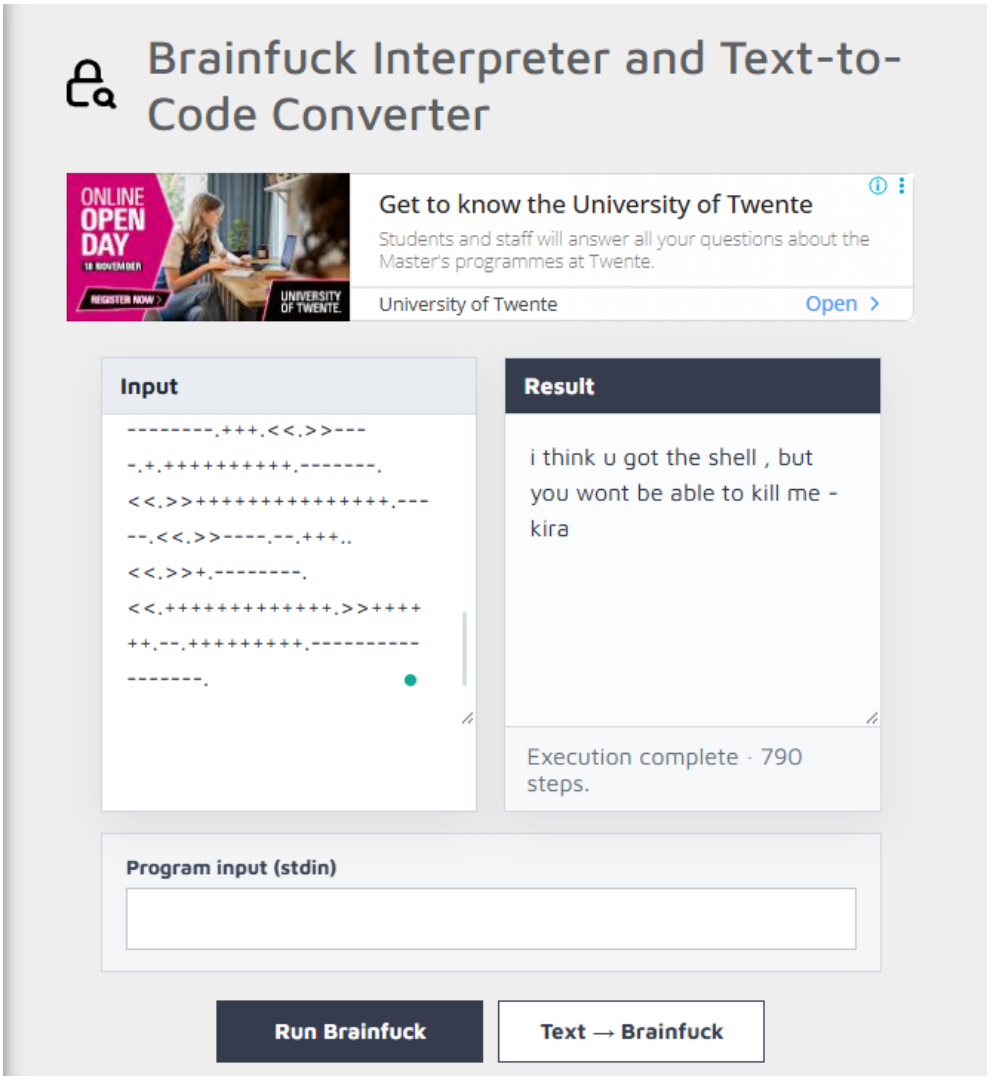


Figure 7: Decoded user.txt message referencing Kira.

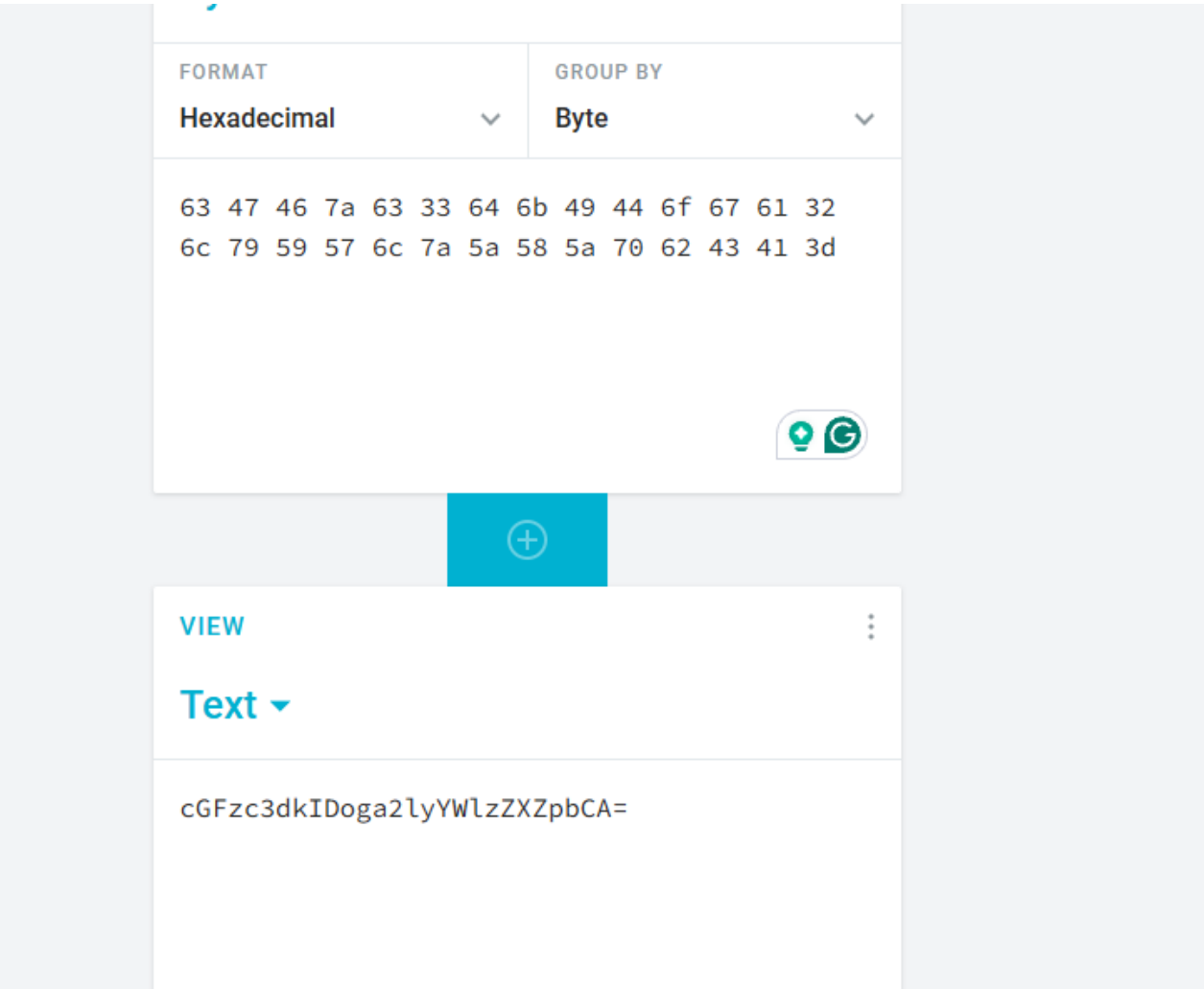


Figure 8: Hexadecimal data converted to text, producing a Base64-encoded clue.



Figure 9: Decoded password clue: passwd : kiraisevil.

4.4 SSH Credential Discovery and Access

The Hydra screenshot shows an SSH password-guessing run against `deathnote.vuln` using candidate usernames and passwords. Its output reports a successful result for username `l` (lowercase L) and password `death4me`. The SSH screenshot shows a successful login to `192.168.56.111` and identifies the host as Debian GNU/Linux x86_64.

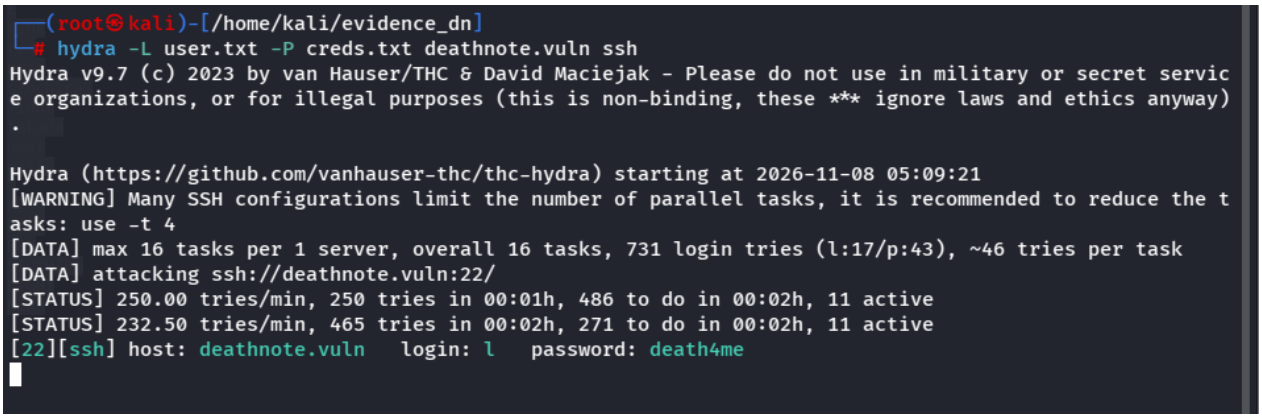


Figure 10: Hydra output showing successful SSH credential discovery for account l (lowercase L).

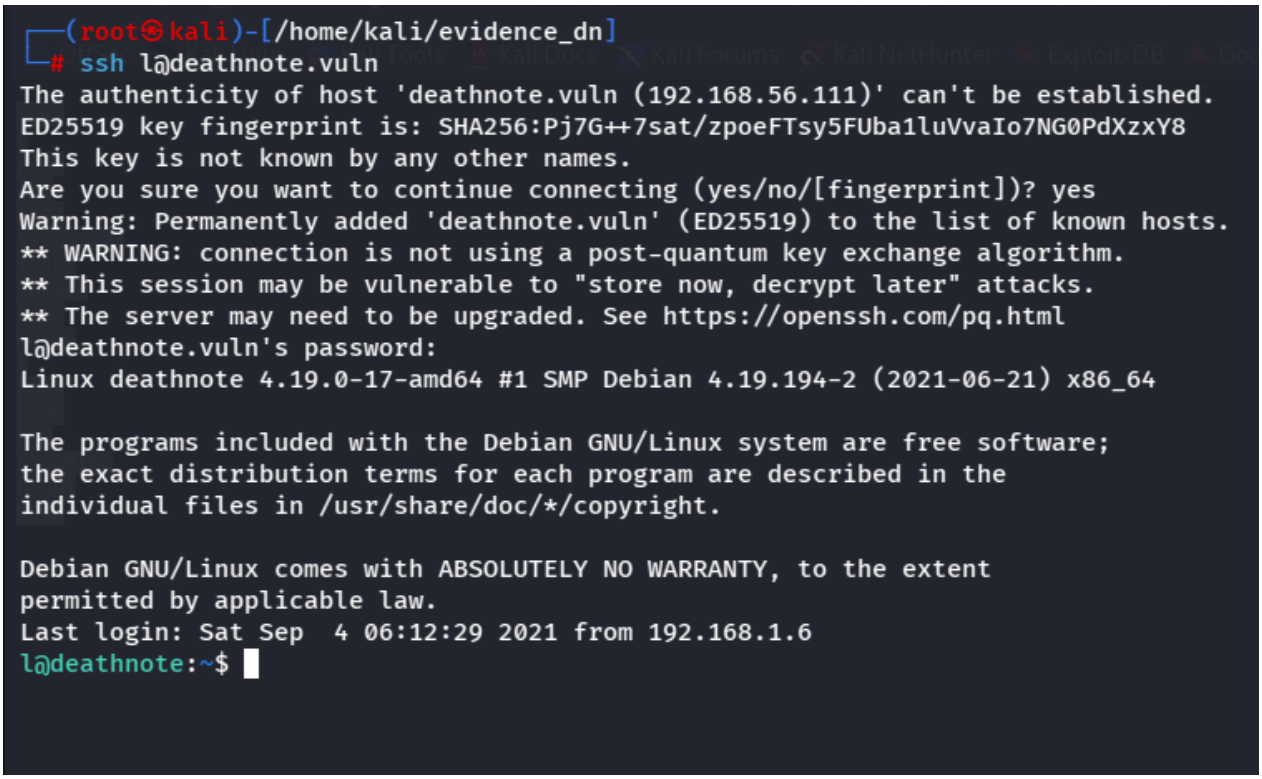


Figure 11: Successful SSH login to deathnote.vuln.

4.5 Local Enumeration and Privilege Escalation

After shell access, the evidence shows a `kira.txt` file containing Base64-encoded text. The decoded instruction references L (`/opt`) and Misa (`/var`). A separate screenshot shows investigation of `/opt/fake-notebook-rule` and `case.wav`.

The final privilege-escalation screenshot displays sudo privileges for `kira`, including `(ALL : ALL) ALL`. The command `sudo su` results in a root shell, after which `/root/root.txt` is accessed.

The available screenshots establish the unrestricted sudo permission as the demonstrated privilege-escalation path. They do not show the exact command sequence used to transition from the initial SSH account to `kira`; that intermediate transition is therefore not asserted.

```
kira@deathnote://home/kira$ cat kira.txt
cGxlYXNlIHByb3RlY3Qgb25lIG9mIHRoZSBmb2xsb3dpbmcgCjEuIEwgKC9vcHQpCjIuIE1pc2EgKC92YXIp
kira@deathnote://home/kira$ echo "cGxlYXNlIHByb3RlY3Qgb25lIG9mIHRoZSBmb2xsb3dpbmcgCjEuIEwgKC9vcHQpCjIuI
E1pc2EgKC92YXIp" | base64 --decode
please protect one of the following
1. L (/opt)
2. Misa (/var)kira@deathnote://home/kira$
```

Figure 12: kira.txt containing a Base64-encoded instruction referencing /opt and /var.

```
l@deathnote:/opt$ cd L
l@deathnote:/opt/L$ ls
fake-notebook-rule  kira-case
l@deathnote:/opt/L$ cd fake-notebook-rule
l@deathnote:/opt/L/fake-notebook-rule$ ls
case.wav  hint
l@deathnote:/opt/L/fake-notebook-rule$ cat case.wav
63 47 46 7a 63 33 64 6b 49 44 6f 67 61 32 6c 79 59 57 6c 7a 5a 58 5a 70 62 43 41 3d
l@deathnote:/opt/L/fake-notebook-rule$
```

Figure 13: case.wav evidence from the investigation of /opt/fake-notebook-rule.

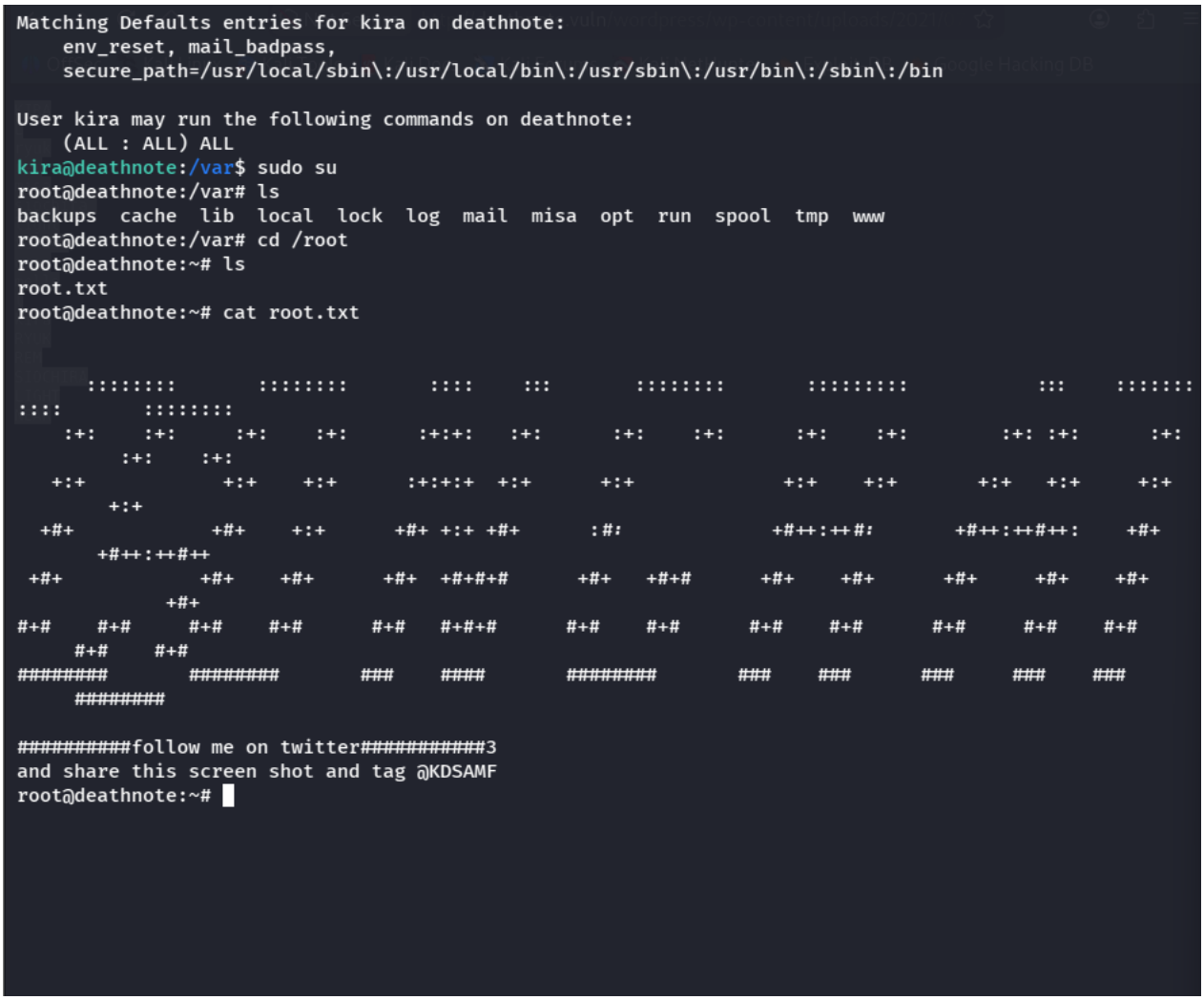


Figure 14: Privilege-escalation evidence showing unrestricted sudo permissions for kira and root access.

5 Key Security Observations

- Publicly accessible WordPress paths and directory listing increased information exposure.
- `robots.txt` and publicly retrievable image content disclosed hints that assisted further enumeration.
- Sensitive clues were stored using reversible encodings. Encoding is not a security control and does not protect secrets.
- SSH password guessing succeeded using a candidate credential list, indicating weak or guessable credentials in the lab environment.
- The `kira` account had unrestricted sudo privileges, allowing immediate escalation to root.

6 Remediation Recommendations

1. **Restrict web directory access:** Disable directory listing, particularly in upload locations, and restrict access to files that are not intended to be public.

2. **Prevent information disclosure:** Review `robots.txt`, publicly accessible media and website content for unintended disclosure. Do not place secrets or sensitive operational hints in public resources.
3. **Protect credentials:** Do not store credentials or sensitive information in reversible encodings such as Base64, hexadecimal or Brainfuck. Use proper secret storage and access controls.
4. **Harden SSH authentication:** Enforce strong, unique credentials; disable password authentication where practical and use key-based authentication with appropriate controls.
5. **Apply least privilege:** Restrict sudo permissions to the minimum required commands. Avoid granting unrestricted `(ALL)` privileges to ordinary accounts.
6. **Maintain software:** Keep WordPress, plugins, Apache and the operating system updated; remove unused components and reduce exposed attack surface.
7. **Monitor activity:** Review authentication attempts, web-server access logs and privilege-escalation activity for suspicious behaviour.

7 Conclusion

The Deathnote machine was compromised through a sequence of information gathering, web and file enumeration, encoded-clue analysis, SSH credential discovery and local privilege escalation. The clearest demonstrated privilege-escalation weakness was the unrestricted sudo permission assigned to `kira`. The exercise illustrates how information disclosures can combine with weak credentials and excessive local privileges to produce full system compromise.

8 Limitations and Evidence Handling

This report is based on the screenshots supplied in the evidence archive. No initial Nmap output, complete terminal transcript, full exploit chronology or independently verified hash of the target image was supplied. Port 80 and SSH port 22 are supported by the screenshots; no additional open ports are claimed. The `root.txt` file was accessed, but its full contents are not reproduced.

> **Assessment boundary**
> This is an intentionally vulnerable training machine, not a production penetration test. Before public release, redact recovered passwords, usernames where appropriate, session material and challenge secrets from the report and evidence images.

END OF REPORT